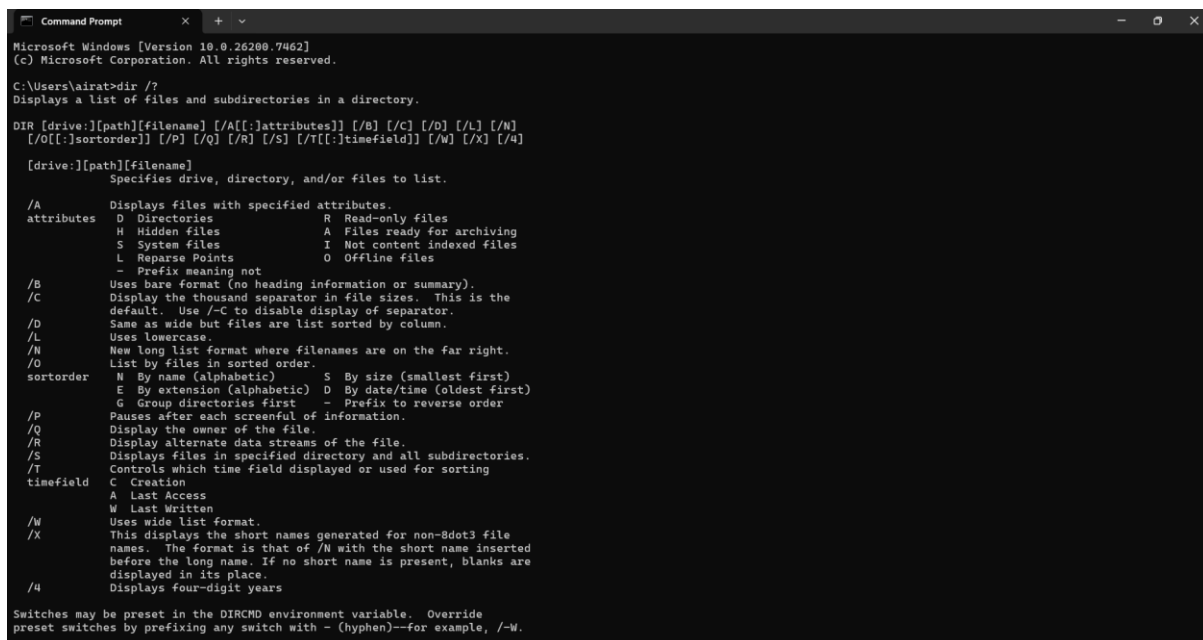


PRACTICAL - 1 John has recently started a new job as system administrator in MNC. He is dealing with computers having different operating systems such as Windows, LINUX and UNIX. How will he be able to interact with the system by using different operating system commands for the following operation?

Output:

1. Display the user manual of any command that we can run on the terminal.



```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>dir /?
Displays a list of files and subdirectories in a directory.

DIR [drive:][path][filename] [/A[[:]attributes]] [/B] [/C] [/O] [/L] [/W]
 [/O[[:]sortorder]] [/P] [/Q] [/R] [/S] [/T[[:]timefield]] [/W] [/X] [/4]

[drive:][path][filename]
    Specifies drive, directory, and/or files to list.

/A      Displays files with specified attributes.
attributes  D Directories          R Read-only files
             H Hidden files        A Files ready for archiving
             S System files        I Not content indexed files
             L Reparse Points      O Offline files
             - Prefix meaning not

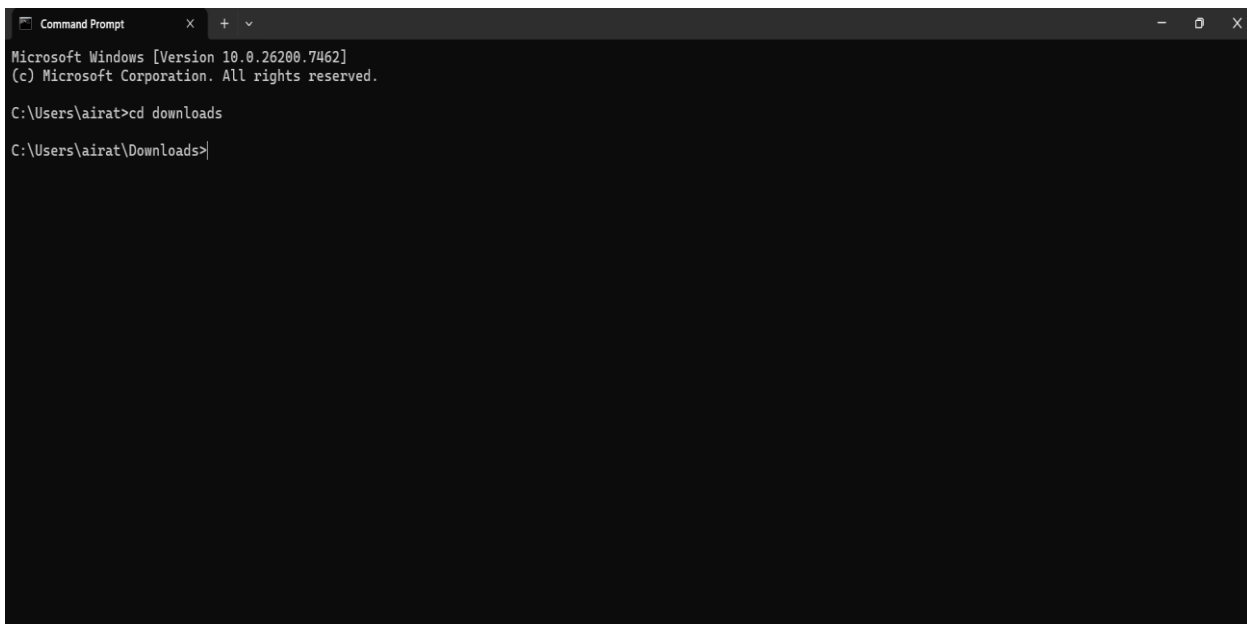
/B      Uses bare format (no heading information or summary).
/C      Display the thousand separator in file sizes. This is the
        default. Use /-C to disable display of separator.
/D      Same as wide but files are list sorted by column.
/L      Uses lowercase.
/N      New long list format where filenames are on the far right.
/O      List by files in sorted order.
sortorder  N By name (alphabetic)    S By size (smallest first)
            E By extension (alphabetic) D By date/time (oldest first)
            G Group directories first - Prefix to reverse order

/P      Pauses after each screenful of information.
/Q      Display the owner of the file.
/R      Display alternate data streams of the file.
/S      Displays files in specified directory and all subdirectories.
/T      Controls which time field displayed or used for sorting
timefield  C Creation
            A Last Access
            W Last Written

/W      Uses wide list format.
/X      This displays the short names generated for non-8dot3 file
        names. The format is that of /N with the short name inserted
        before the long name. If no short name is present, blanks are
        displayed in its place.
/4      Displays four-digit years

Switches may be preset in the DIRCMD environment variable. Override
preset switches by prefixing any switch with - (hyphen)--for example, /-W.
```

2. Change the current working directory of the user



```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>cd downloads
C:\Users\airat\Downloads>
```

3. List all the contents in the current working directory.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>dir
Volume in drive C is Windows
Volume Serial Number is 4A3B-E840

Directory of C:\Users\airat

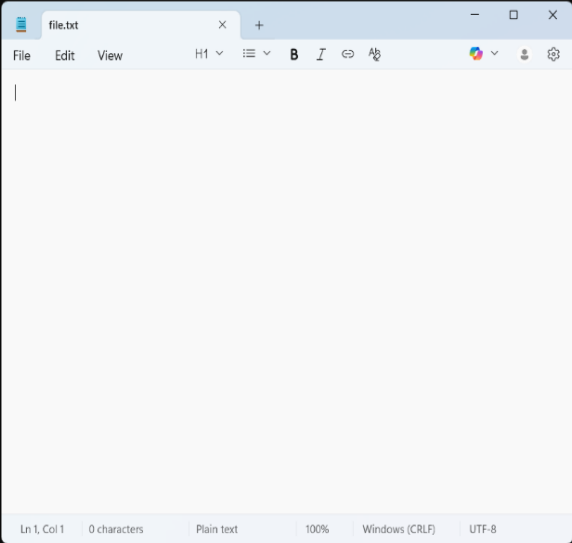
10-01-2026  23:42    <DIR>        .
28-10-2025  13:08    <DIR>        ..
06-10-2025  22:15    <DIR>        .arduinoIDE
04-10-2025  01:25    <DIR>        .vscode
04-10-2025  00:52             1,844 AMDRM_Install.log
04-10-2025  01:07    <DIR>        ansel
28-10-2025  13:12    <DIR>        Contacts
03-10-2025  19:10    <DIR>        Documents
12-01-2026  21:13    <DIR>        Downloads
28-10-2025  13:12    <DIR>        Favorites
28-10-2025  13:12    <DIR>        Links
28-10-2025  13:12    <DIR>        Music
12-01-2026  20:53    <DIR>        OneDrive
10-01-2026  22:49    <DIR>        Saved Games
28-10-2025  13:13    <DIR>        Searches
28-10-2025  13:12    <DIR>        Videos
               1 File(s)              1,844 bytes
               15 Dir(s)  15,895,101,440 bytes free

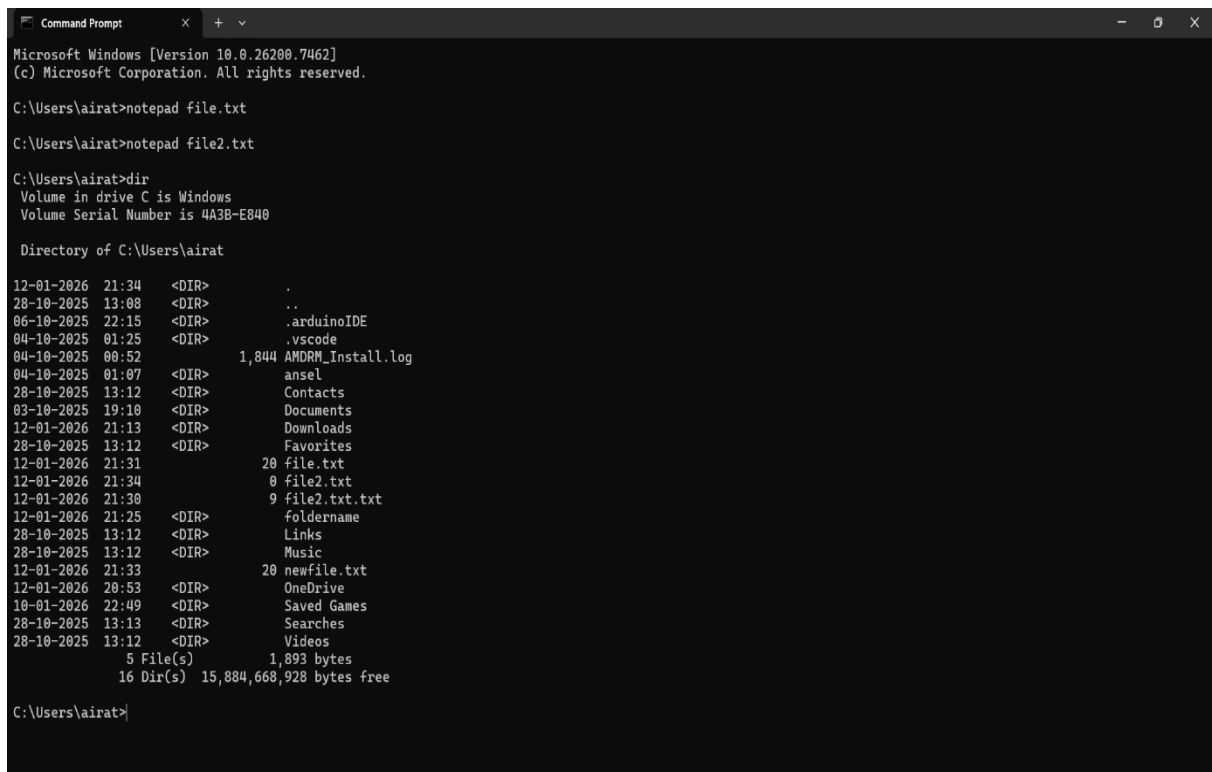
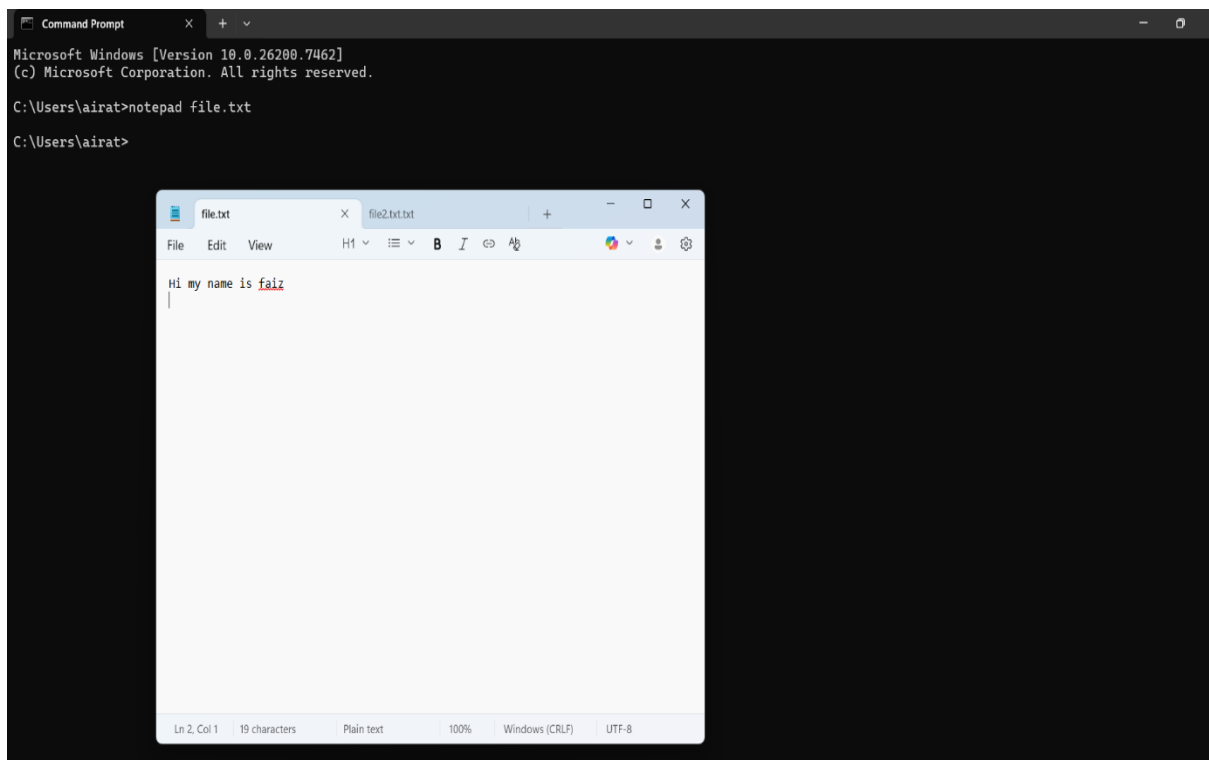
C:\Users\airat>
```

4. Read modify or concatenate files

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>notepad file.txt
C:\Users\airat>notepad file.txt
C:\Users\airat>
```





5. Create a new directory under any directory

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>mkdir TestFolder

C:\Users\airat>dir
Volume in drive C is Windows
Volume Serial Number is 4A3B-E840

Directory of C:\Users\airat

12-01-2026  21:41    <DIR>          .
28-10-2025  13:08    <DIR>          ..
06-10-2025  22:15    <DIR>          .arduinoIDE
04-10-2025  01:25    <DIR>          .vscode
04-10-2025  00:52             1,844 AMDRM_Install.log
04-10-2025  01:07    <DIR>          ansel
28-10-2025  13:12    <DIR>          Contacts
03-10-2025  19:10    <DIR>          Documents
12-01-2026  21:13    <DIR>          Downloads
12-01-2026  21:40    <DIR>          Faiz
28-10-2025  13:12    <DIR>          Favorites
12-01-2026  21:31             20 file.txt
12-01-2026  21:34              0 file2.txt
12-01-2026  21:38              9 file2.txt.txt
12-01-2026  21:25    <DIR>          foldername
28-10-2025  13:12    <DIR>          Links
28-10-2025  13:12    <DIR>          Music
12-01-2026  21:33             20 newfile.txt
12-01-2026  20:53    <DIR>          OneDrive
10-01-2026  22:49    <DIR>          Saved Games
28-10-2025  13:13    <DIR>          Searches
12-01-2026  21:41    <DIR>          TestFolder
28-10-2025  13:12    <DIR>          Videos
           5 File(s)              1,893 bytes
           18 Dir(s)  15,878,746,112 bytes free

C:\Users\airat>
```

6. Display the current working directory of the terminal

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>echo %cd%
C:\Users\airat
C:\Users\airat>
```

7. Write its arguments to standard output

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>echo Welcome to System Administration Lab
Welcome to System Administration Lab

C:\Users\airat>
```

8. Remove a file

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>notepad delete_me.txt

C:\Users\airat>dir
Volume in drive C is Windows
Volume Serial Number is 4A3B-E840

Directory of C:\Users\airat

12-01-2026 22:03 <DIR> .
28-10-2025 13:08 <DIR> ..
06-10-2025 22:15 <DIR> .arduinoIDE
04-10-2025 01:25 <DIR> .vscode
04-10-2025 00:52 <DIR> 1,844 AMDRM_Install.log
04-10-2025 01:07 <DIR> ansel
28-10-2025 13:12 <DIR> Contacts
12-01-2026 22:03 <DIR> 4 delete_me.txt
03-10-2025 19:10 <DIR> Documents
12-01-2026 21:13 <DIR> Downloads
12-01-2026 21:40 <DIR> Faiz
28-10-2025 13:12 <DIR> Favorites
12-01-2026 21:31 <DIR> 20 file.txt
12-01-2026 21:34 <DIR> 0 file2.txt
12-01-2026 21:30 <DIR> 9 file2.txt.txt
12-01-2026 21:25 <DIR> foldername
28-10-2025 13:12 <DIR> Links
28-10-2025 13:12 <DIR> Music
12-01-2026 21:33 <DIR> 20 newfile.txt
12-01-2026 20:53 <DIR> OneDrive
10-01-2026 22:49 <DIR> Saved Games
28-10-2025 13:13 <DIR> Searches
12-01-2026 21:41 <DIR> TestFolder
28-10-2025 13:12 <DIR> Videos
        6 File(s)          1,897 bytes
       18 Dir(s) 15,940,702,208 bytes free

C:\Users\airat>
```

9. Delete a directory.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>mkdir RemoveFolder
C:\Users\airat>rmdir RemoveFolder
C:\Users\airat>dir
Volume in drive C is Windows
Volume Serial Number is 4A3B-E840

Directory of C:\Users\airat

12-01-2026 22:06 <DIR>      .
28-10-2025 13:08 <DIR>      ..
06-10-2025 22:15 <DIR>      .arduinoIDE
04-10-2025 01:25 <DIR>      .vscode
04-10-2025 00:52 <DIR>      1,844 AMDRM_Install.log
04-10-2025 01:07 <DIR>      ansel
28-10-2025 13:12 <DIR>      Contacts
12-01-2026 22:03 <DIR>      4 delete_me.txt
03-10-2025 19:10 <DIR>      Documents
12-01-2026 21:13 <DIR>      Downloads
12-01-2026 21:40 <DIR>      Faiz
28-10-2025 13:12 <DIR>      Favorites
12-01-2026 21:31 <DIR>      20 file.txt
12-01-2026 21:34 <DIR>      0 file2.txt
12-01-2026 21:30 <DIR>      9 file2.txt.txt
12-01-2026 21:25 <DIR>      foldername
28-10-2025 13:12 <DIR>      Links
28-10-2025 13:12 <DIR>      Music
12-01-2026 21:33 <DIR>      20 newfile.txt
12-01-2026 20:53 <DIR>      OneDrive
10-01-2026 22:49 <DIR>      Saved Games
28-10-2025 13:13 <DIR>      Searches
12-01-2026 21:41 <DIR>      TestFolder
28-10-2025 13:12 <DIR>      Videos
        6 File(s)          1,897 bytes
       18 Dir(s) 15,914,336,256 bytes free

C:\Users\airat>
```

10. Copy a file or directory

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>notepad copyfile.txt
C:\Users\airat>copyfile.txt TextFolder\
C:\Users\airat>copy copyfile.txt TextFolder\
The system cannot find the path specified.
0 file(s) copied.

C:\Users\airat>dir TestFolder
Volume in drive C is Windows
Volume Serial Number is 4A3B-E840

Directory of C:\Users\airat\TestFolder

12-01-2026 21:41 <DIR>      .
12-01-2026 22:07 <DIR>      ..
        0 File(s)          0 bytes
        2 Dir(s) 15,915,814,912 bytes free

C:\Users\airat>
```

11. Switch to root-user

```
Select Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>runAs

RUNAS USAGE:

RUNAS [ [/noprofile | /profile] [/env] [/savecred | /netonly] ]
        /user:<UserName> program

RUNAS [ [/noprofile | /profile] [/env] [/savecred] ]
        /smartcard [/user:<UserName>] program

RUNAS [ [/machine:<MachineType>] ] /trustlevel:<Trustlevel> program

        /noprofile      specifies that the user's profile should not be loaded.
                        This causes the application to load more quickly, but
                        can cause some applications to malfunction.
        /profile        specifies that the user's profile should be loaded.
                        This is the default.
        /env            to use current environment instead of user's.
        /netonly        use if the credentials specified are for remote
                        access only.
        /savecred       to use credentials previously saved by the user.
        /smartcard      use if the credentials are to be supplied from a
                        smartcard.
        /user           <UserName> should be in form USER@DOMAIN or DOMAIN\USER
        /showtrustlevels displays the trust levels that can be used as arguments
                        to /trustlevel.
        /trustlevel     <level> should be one of levels enumerated
                        in /showtrustlevels.
        /machine        specifies the machine architecture of the process.
                        <MachineType> should be one of x86|amd64|arm|arm64.
        program         command line for EXE. See below for examples

Examples:
> runas /noprofile /user:mymachine\administrator cmd
> runas /profile /env /user:mydomain\admin "mmc %windir%\system32\dsa.msc"
> runas /env /user:user@domain.microsoft.com "notepad \"my file.txt\""

NOTE: Enter user's password only when prompted.
NOTE: /profile is not compatible with /netonly.
NOTE: /savecred is not compatible with /smartcard.

C:\Windows\System32>
```

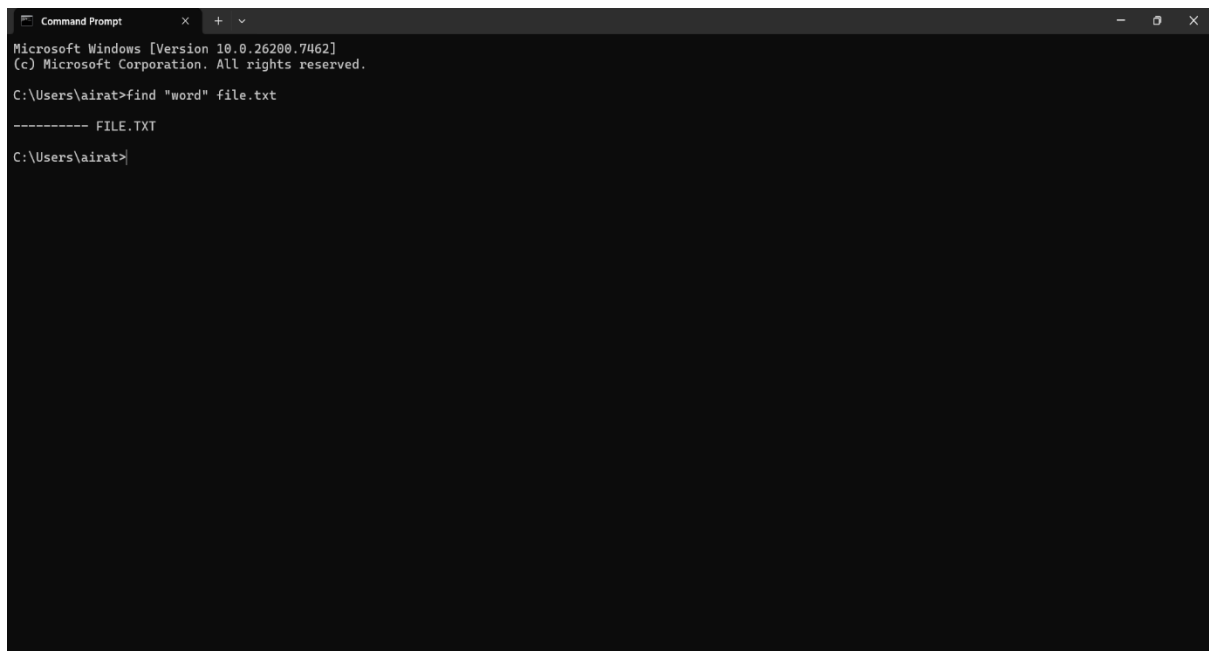
12. Moves files and directories from one directory to another

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>move file.txt D:\Backup\
The system cannot find the path specified.
0 file(s) moved.

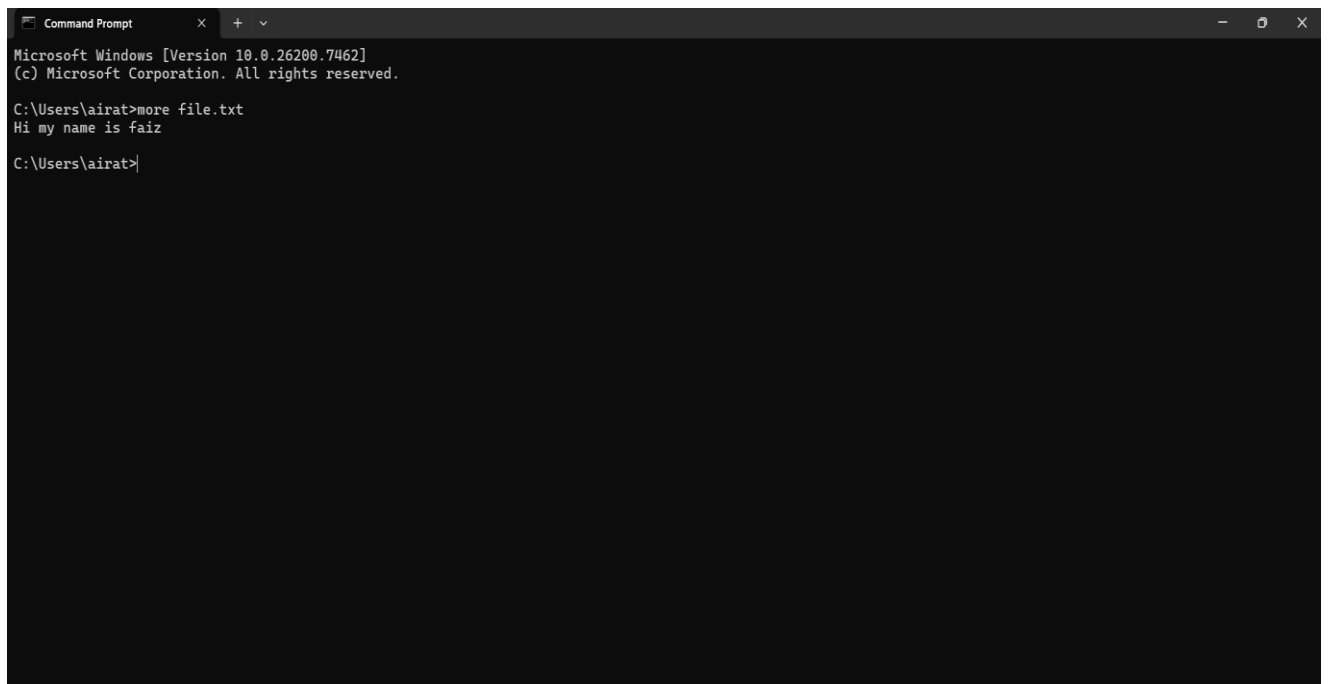
C:\Users\airat>|
```

13. Search for a particular string/word in a text file



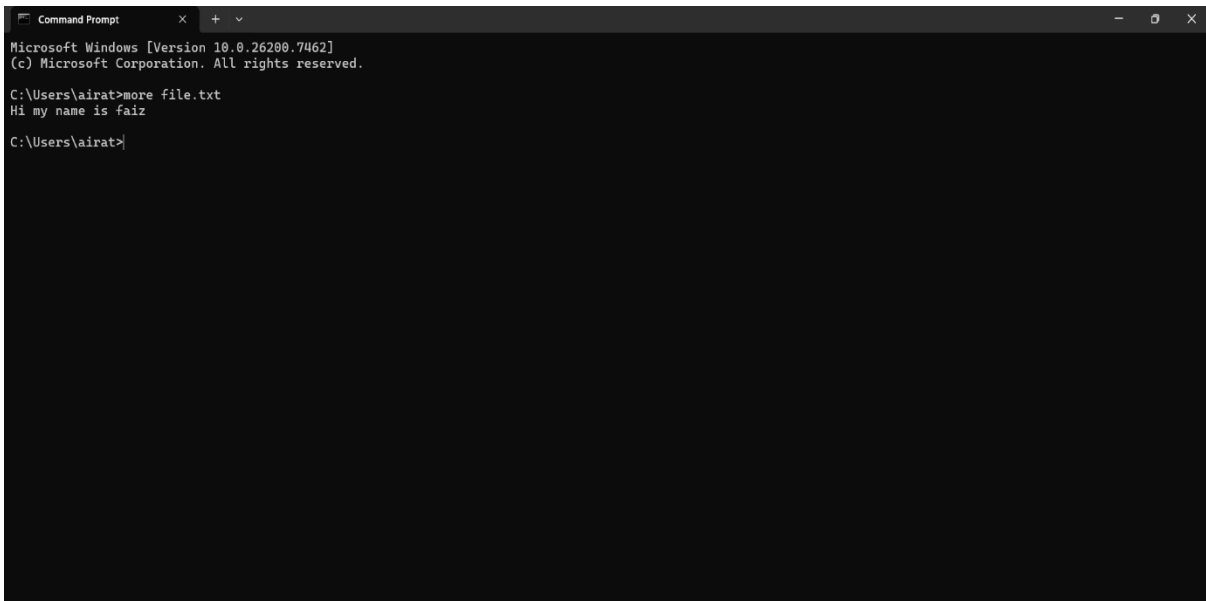
```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.
C:\Users\airat>find "word" file.txt
----- FILE.TXT
C:\Users\airat>
```

14. Print the top N number of data of the given input



```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.
C:\Users\airat>more file.txt
Hi my name is faiz
C:\Users\airat>
```


15. Print the last N number of data of the given input

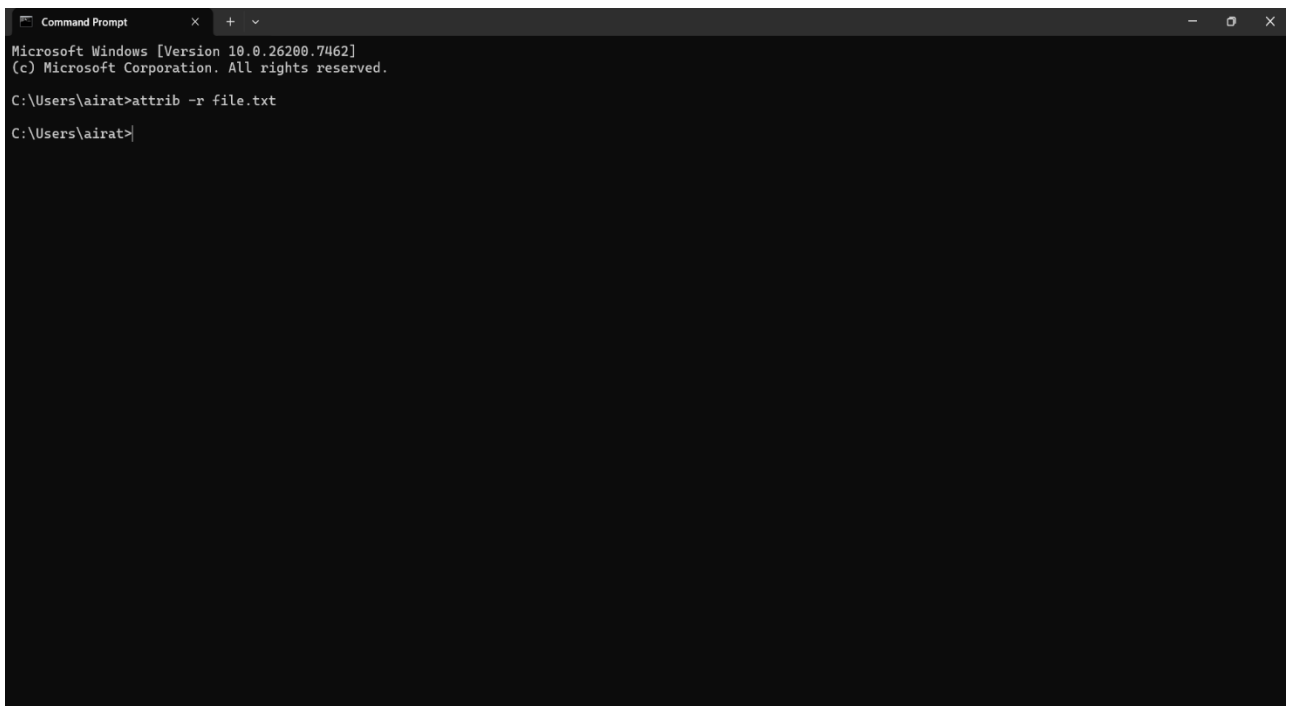


```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>more file.txt
Hi my name is faiz

C:\Users\airat>
```

16. Remove read permission from owner of the file

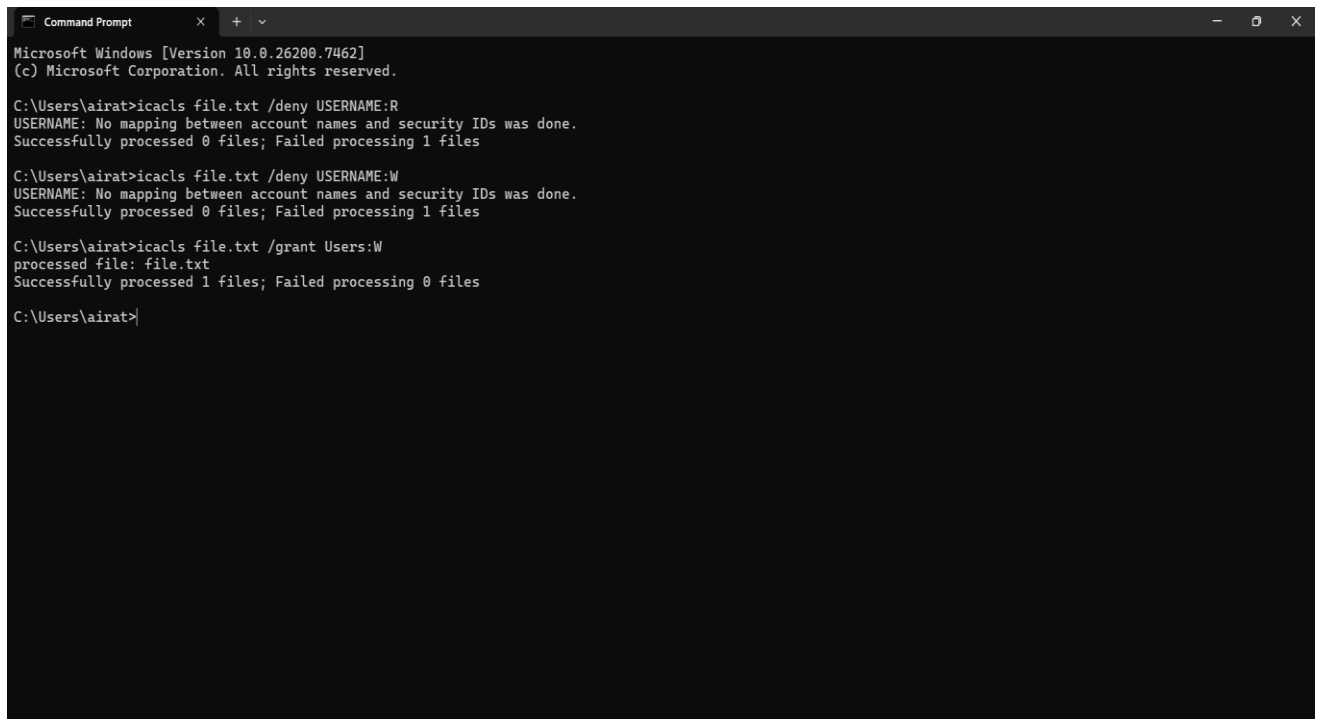


```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>attrib -r file.txt

C:\Users\airat>
```

17. Add read permission to owner plus re move write permission and
add write permission to group



```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

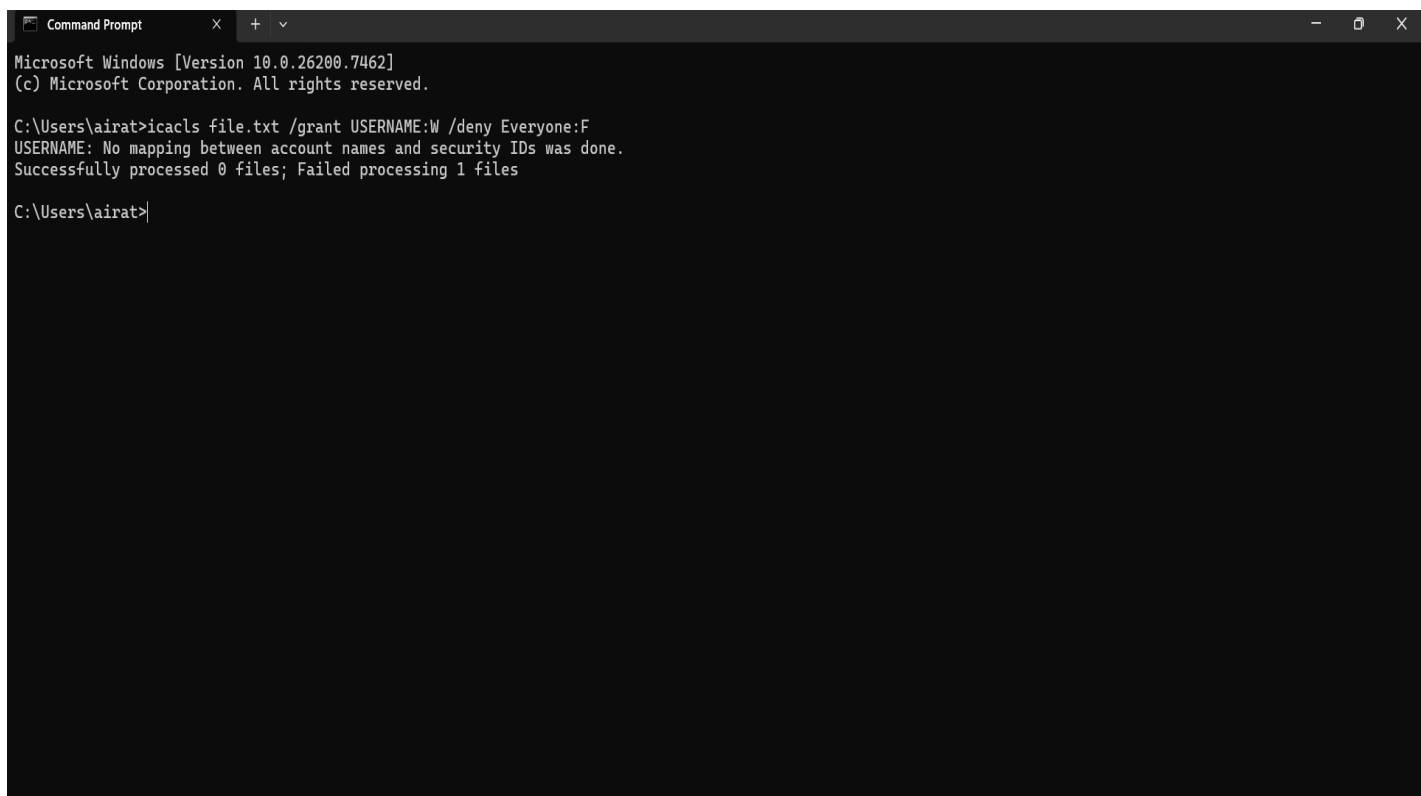
C:\Users\airat>icacls file.txt /deny USERNAME:R
USERNAME: No mapping between account names and security IDs was done.
Successfully processed 0 files; Failed processing 1 files

C:\Users\airat>icacls file.txt /deny USERNAME:W
USERNAME: No mapping between account names and security IDs was done.
Successfully processed 0 files; Failed processing 1 files

C:\Users\airat>icacls file.txt /grant Users:W
processed file: file.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\airat>
```

18. Add write permission to owner no permission to others

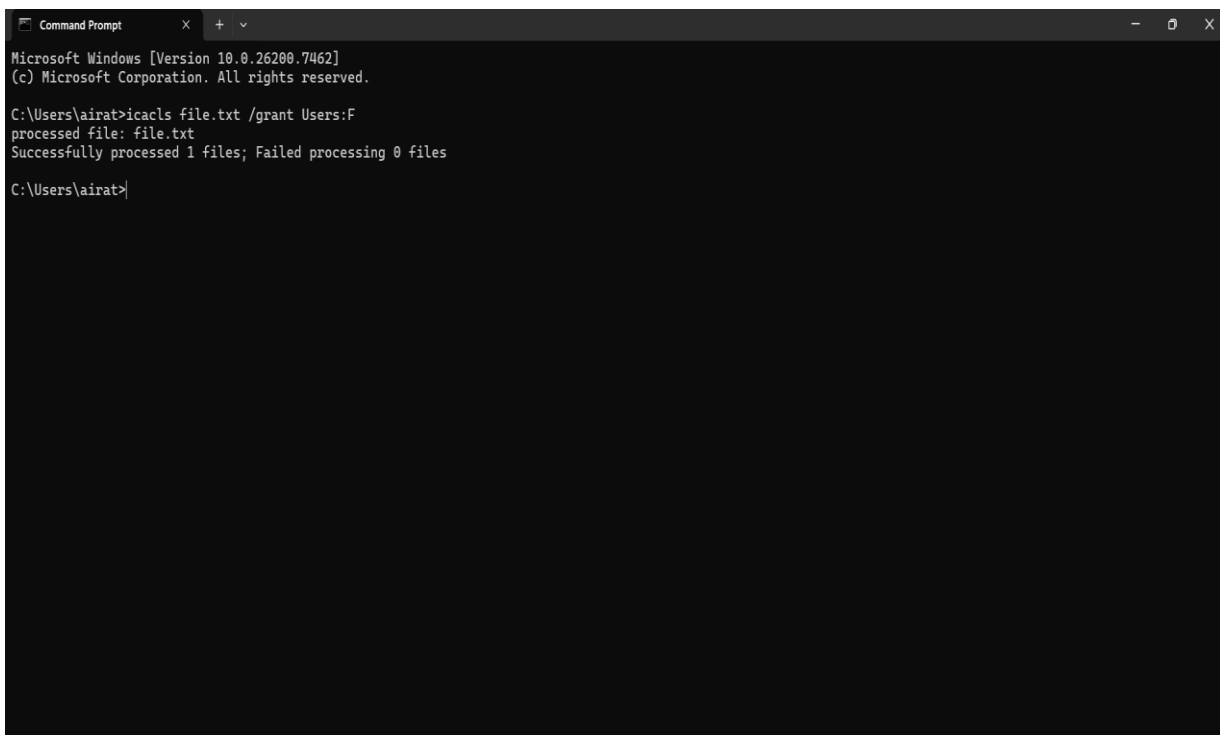


```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>icacls file.txt /grant USERNAME:W /deny Everyone:F
USERNAME: No mapping between account names and security IDs was done.
Successfully processed 0 files; Failed processing 1 files

C:\Users\airat>
```

19. Assign permission to users

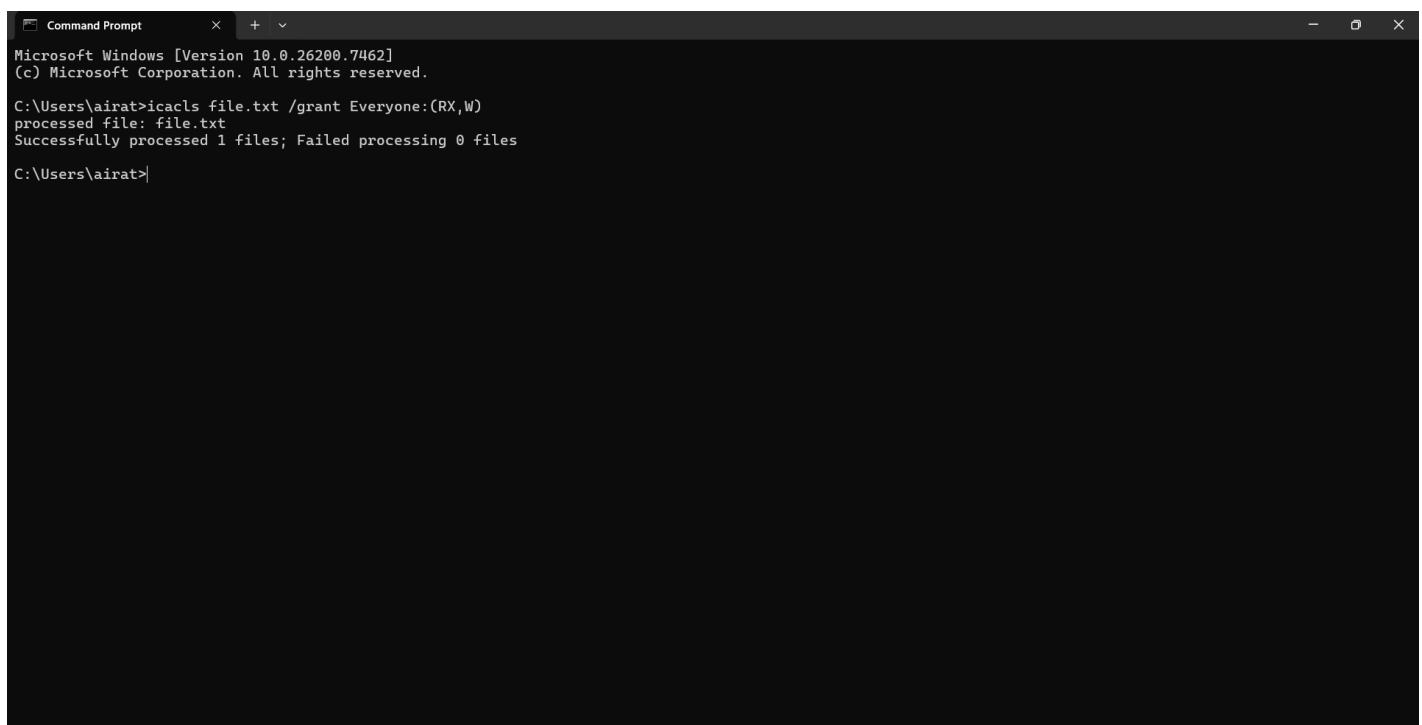


```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>icacls file.txt /grant Users:F
processed file: file.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\airat>
```

20. Assign R/W/X to others

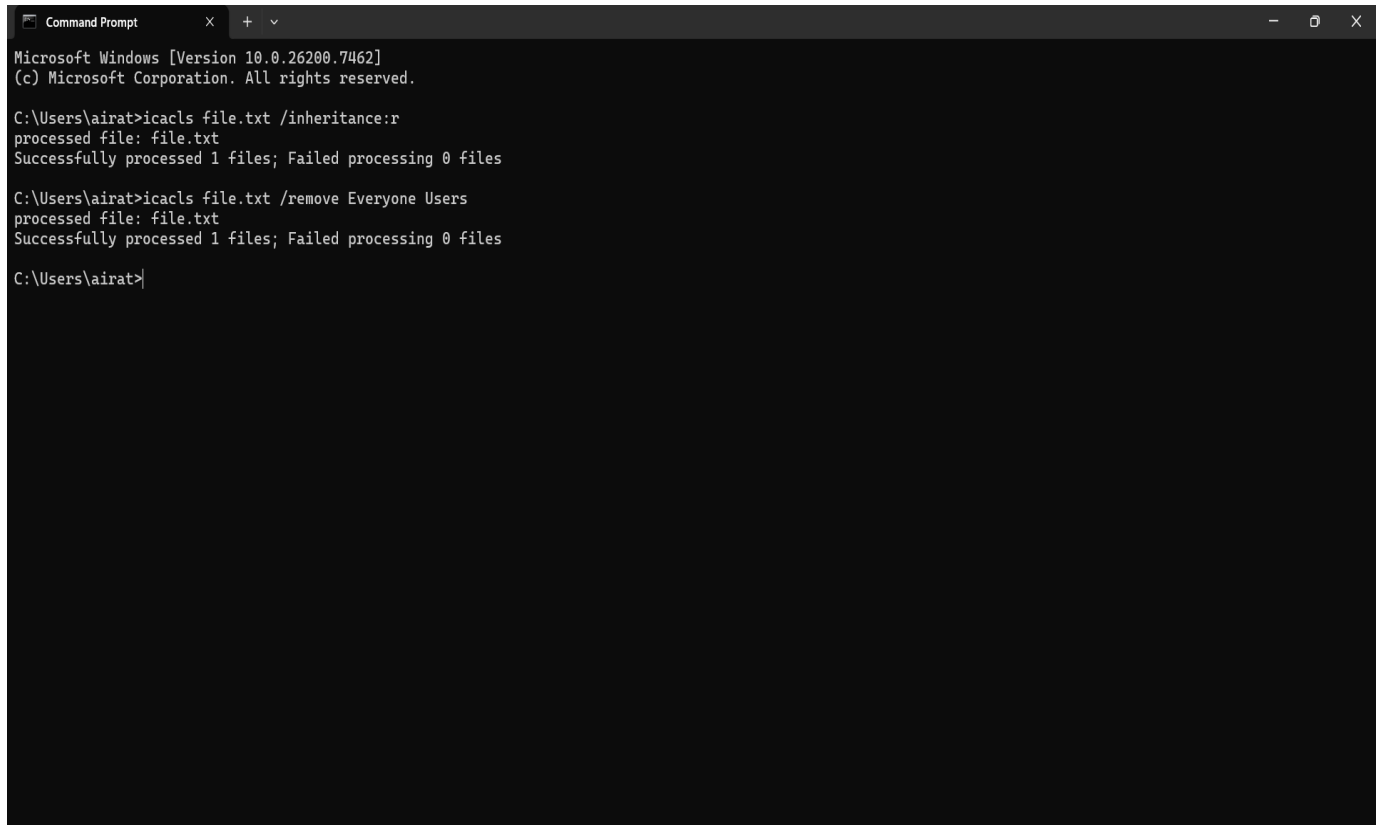


```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>icacls file.txt /grant Everyone:(RX,W)
processed file: file.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\airat>
```

21. Remove all permission from all users



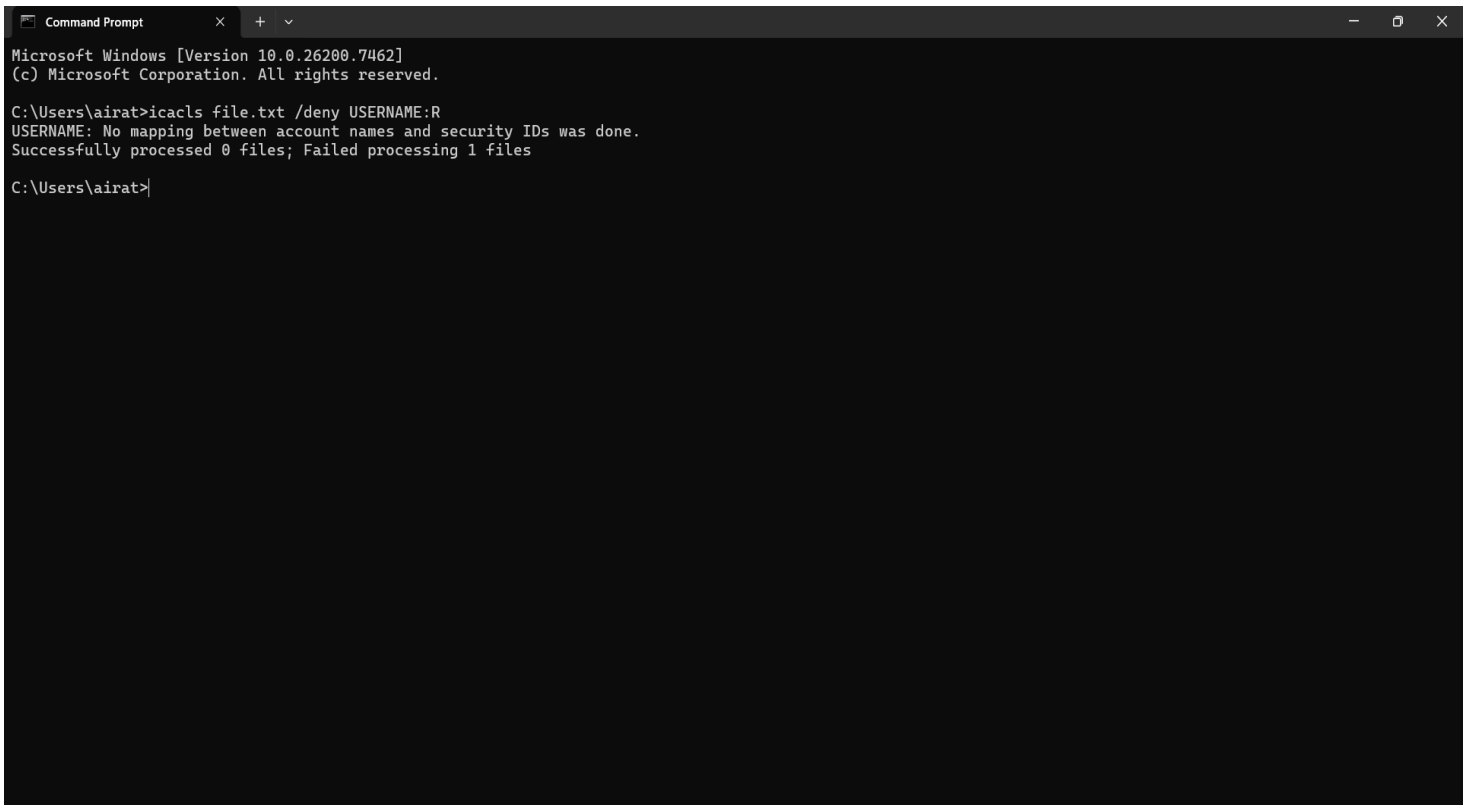
```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>icacls file.txt /inheritance:r
processed file: file.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\airat>icacls file.txt /remove Everyone Users
processed file: file.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\airat>
```

22. Remove read permission from user using absolute mode

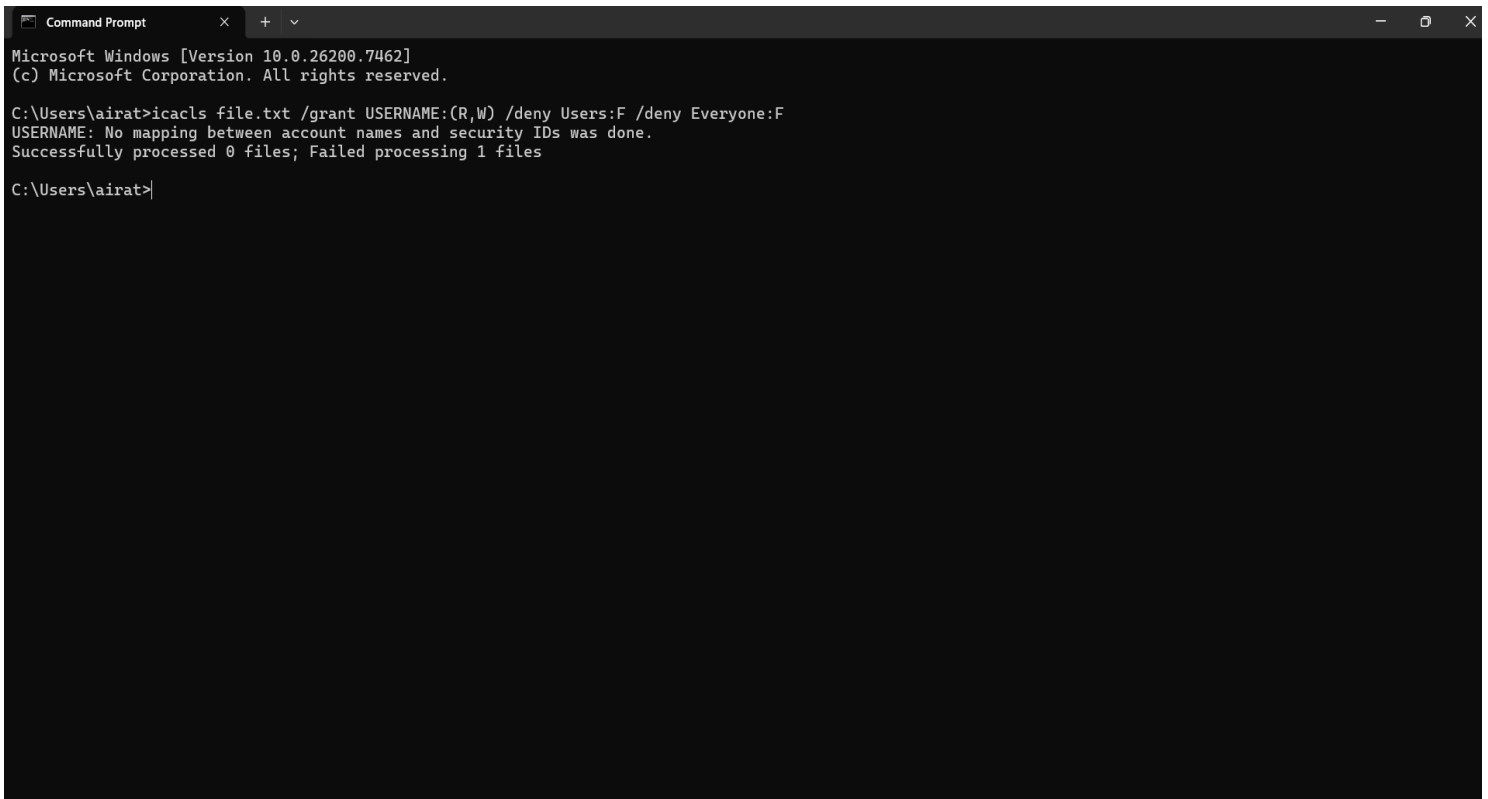


```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>icacls file.txt /deny USERNAME:R
USERNAME: No mapping between account names and security IDs was done.
Successfully processed 0 files; Failed processing 1 files

C:\Users\airat>
```

23. Add R/w permission to owner no permission to group and no permission to other using absolute mode

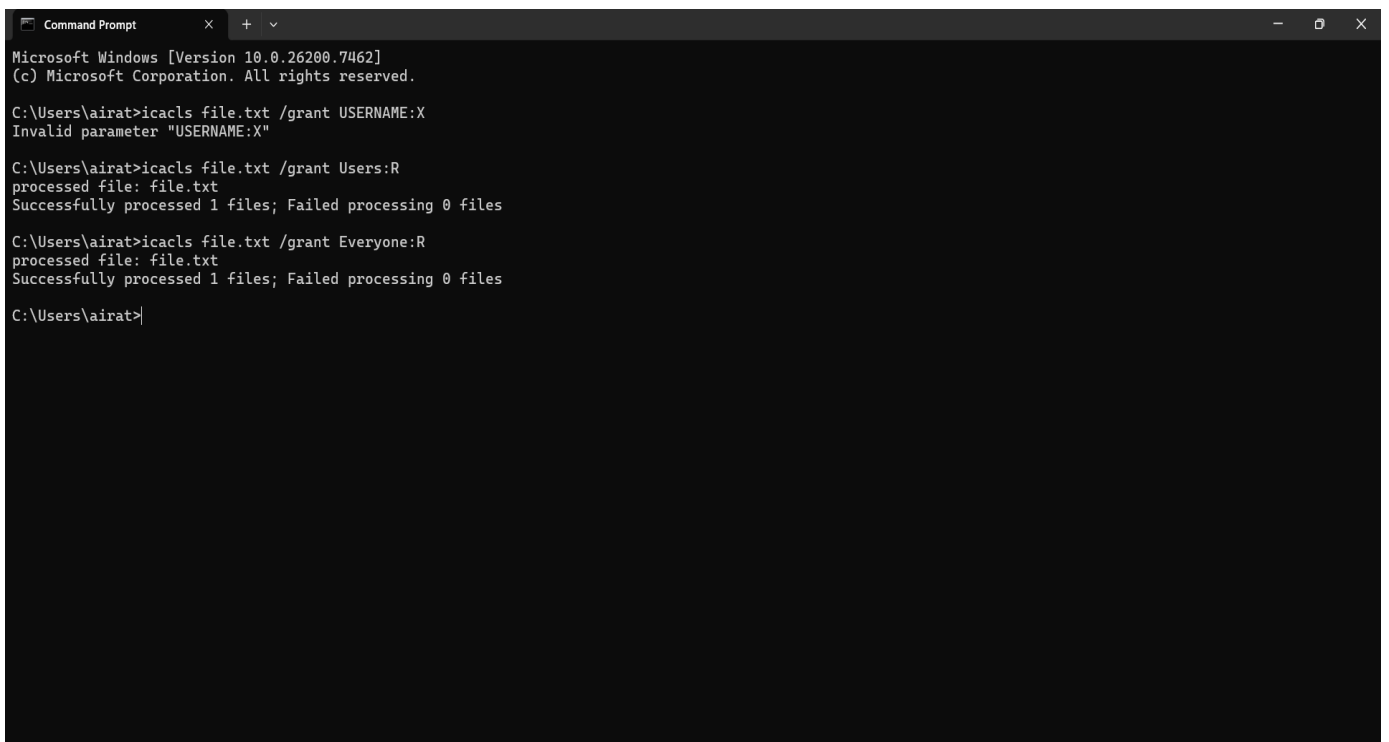


```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>icacls file.txt /grant USERNAME:(R,W) /deny Users:F /deny Everyone:F
USERNAME: No mapping between account names and security IDs was done.
Successfully processed 0 files; Failed processing 1 files

C:\Users\airat>
```

24. Add execute permission to owner, read permission to group and other



```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

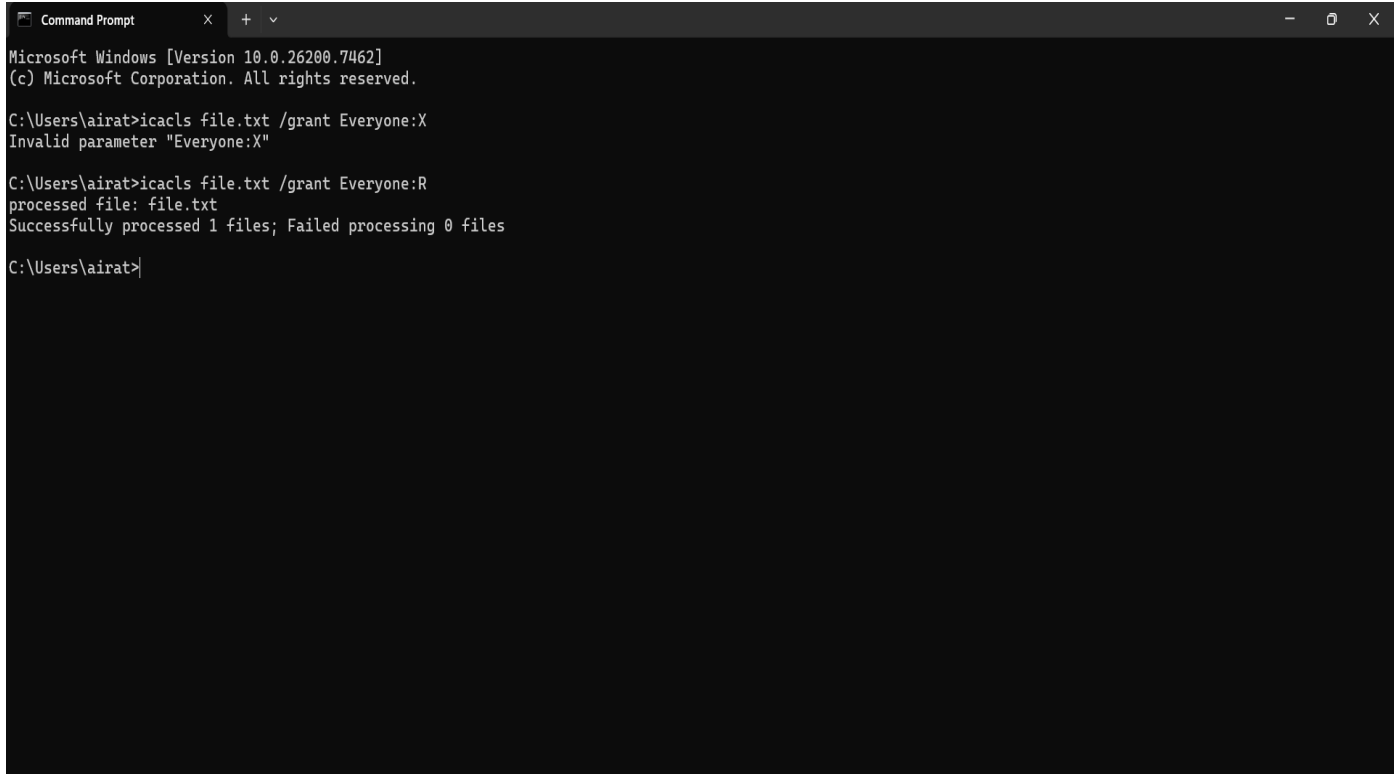
C:\Users\airat>icacls file.txt /grant USERNAME:X
Invalid parameter "USERNAME:X"

C:\Users\airat>icacls file.txt /grant Users:R
processed file: file.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\airat>icacls file.txt /grant Everyone:R
processed file: file.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\airat>
```

25. Add execute permission to all user



```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\airat>icacls file.txt /grant Everyone:X
Invalid parameter "Everyone:X"

C:\Users\airat>icacls file.txt /grant Everyone:R
processed file: file.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\airat>
```